Q16 GENUS-FOUR GHS FRONTIER

Exact 66-curve census, evaluated subgroup maps, checked genus bound, and same-target attack costs | 28 October 2026

Finding. All 66 ordinary curves over F_(16^3) outside the K-isogeny closure of three earlier direct genus-three V4 screens have an evaluated, nonzero prime-subgroup transfer to a genus-four hyperelliptic Jacobian over F_16. A complete standard GHS parameter sweep returned no genus-three curve for any of the 66 equations. Full relation solves on three representatives were slower than rho.

Exact finite cohort

Let K = F_2[w]/(w^12+w^7+w^6+w^5+w^3+w+1), k = F_16, and E_b: y^2 + xy = x^3 + w^5 x^2 + b. Each b is a polynomial-basis bit mask. The source certificate in the repository lists every mask and proves the three-screen isogeny exclusion.

#E(K)	Models	Prime r	b of measured source	c=1 Jacobian order(s)
3974	12	1987	306	55,636 (12)
3978	30	17	11	39,780 (6), 71,604 (24)
4178	24	2089	31	75,204 (24)

Exhaustive parameter result: for each equation, all 4,095 nonzero c in K were evaluated by Magma 2.30-1: 30 genus four, 450 genus six, 3,585 genus seven, and 30 intrinsic rejections. Across 66 equations this accounts for 270,270 pairs, with zero genus-three outputs. The bound is exact for this documented intrinsic and parameterization; it is not a classification of all covers or isogenies.

Subgroup map: Magma's WeilDescent(E_b,k,1) returned C_b and h_b. For every b, the retained script checks genus(C_b)=4, G_b nonzero of order r, h_b(G_b) nonzero of order r, and h_b(7G_b)=7h_b(G_b). The exact G_b and reduced Mumford images are in the 66-model receipts.

Necessary genus bound for any transfer

For every one of the 66 source models, the Weil-restriction Frobenius polynomial is X^6 - tX^3 + 4096, with t in {123, 119, -81}. Sage Rabin tests prove these polynomials irreducible modulo 11, 11, and 5 respectively. By the finite-field isogeny theorem, each restriction is a simple threefold. Any nonzero algebraic transfer from it to a Jacobian requires target genus at least three.

Independently, if a prime-order subgroup r maps nontrivially into J(C)(F_q), then r divides #J(C)(F_q). The Weil bound implies r <= (sqrt(q)+1)^(2g). At q=16, genus at most two gives #J <= 5^4 = 625; hence the r=1,987 and r=2,089 subgroups require g >= 3 for any subgroup-preserving transfer. GenusBound.lean checks the finite divisibility contradiction. The Weil bound, Lagrange's theorem, and the coordinate checks are explicit external inputs.

End-to-end representative comparison

#E (b)	Relations	GHS steps	GHS CPU	Rho mean steps	Rho mean CPU	GHS/rho
3974 (306)	13	1,188	0.22 s	75.11	0.00102 s	216x
3978 (11)	17	4,172	0.33 s	6.81	0.00010 s	3300x
4178 (31)	17	479	0.20 s	72.15	0.00101 s	198x

All three relation matrices recover 7 and verify it in the source and target groups. The source rho column uses 1,000 successful same-target trials with degenerate-collision restarts. Magma CPU has 0.01-second reporting granularity; ratios are finite-size same-backend observations. The remaining 63 full attack costs are unmeasured.

Prior work and claim boundary

Binary cubic GHS genus-three/genus-four outcomes and Magma's transfer intrinsic are established prior art. All three Weil-restriction isogeny classes already have catalogued genus-three hyperelliptic Jacobians. The order-3,978 class also appears in earlier repository work. Thus this is a new exact finite census and map certificate, not an established new family, new method, or faster attack. An evaluated source-to-genus-three isogeny remains the main open route.

Sources: Gaudry-Hess-Smart, Weil Descent on Elliptic Curves (kantmath.de/publications/papers/weil_descent.pdf); Magma handbook (magma.maths.usyd.edu.au/magma/handbook/text/1561); Murai et al., SCIS 2022 abstract; LMFDB classes 3.16.a_a_aet, 3.16.a_a_aep, 3.16.a_a_dd. Executable scripts, full receipts, Lean theorem, and 66-row summary are in the companion research directory.